

Internal Assessment Question Paper – 1

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Artificial Intelligence and Data Science
Term: May-Aug 2023

Programme: B.E
30

CIE: Test 1
AD43/A143

Sem: IV

LIS)

Subject: Design and Analysis of Algorithms

Crédits: 3:0:0:0

Date: 12/12/2022

Time: 2:00 to 3:00

Max Marks:

Subject Code:

Portions for Test: (L1-

Instructions to Candidates:

1. Question 1 is Compulsory. Answer any one question from 2 and 3.
2. Each Question carries 15M. Write figures wherever necessary.
3. Mobiles/Smart Gadgets are restricted.

Sl#	Question	Marks	Bloom's Level	CO Mapping																																								
1	a. Find the stable matching for the following Men set (A, B, C & D) and Women's Set (V, W, X & Y). <table border="1"> <tr><td>A</td><td>V</td><td>W</td><td>X</td><td>Y</td></tr> <tr><td>B</td><td>W</td><td>X</td><td>Y</td><td>V</td></tr> <tr><td>C</td><td>X</td><td>Y</td><td>W</td><td>V</td></tr> <tr><td>D</td><td>V</td><td>W</td><td>X</td><td>Y</td></tr> </table> <table border="1"> <tr><td>V</td><td>D</td><td>C</td><td>B</td><td>A</td></tr> <tr><td>W</td><td>A</td><td>D</td><td>C</td><td>B</td></tr> <tr><td>X</td><td>B</td><td>A</td><td>D</td><td>C</td></tr> <tr><td>Y</td><td>A</td><td>D</td><td>C</td><td>D</td></tr> </table> b. i. What is the time complexity of the following code? <pre> int a = 0, i = N; while (i > 0) { a += i; i /= 2; } </pre> ii. Find a maximum independent set from the above graph. c. Construct the DFS Traversal tree for the following graph with "A" as the source vertex.	A	V	W	X	Y	B	W	X	Y	V	C	X	Y	W	V	D	V	W	X	Y	V	D	C	B	A	W	A	D	C	B	X	B	A	D	C	Y	A	D	C	D	06	Apply	CO1
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		04	Apply	CO1																																								
		05	Apply	CO2																																								
2	a. Define the following with respect to Gale-Shapley algorithm i. Perfect Matching ii. Stable Matching iii. Instability iv. Best Valid Partner b. Prove the following: i. $T(n) = 14n^2 + 6n + 2$ is $O(n^3)$ ii. $T(n) = 14n^2 + 7n + 5$ is $\Omega(n)$ c. Construct a proof for "If f is a polynomial of degree d, in which the coefficient a_d is positive. Then $f = O(n^d)$".	04	Understand	CO1																																								
		06	Apply	CO1																																								
		05	Apply	CO2																																								
		05	Apply	CO1																																								
3	a. Prove the following with respect to Stable Matching Algorithm i. If m is free at some point in the execution of the algorithm, then there is a woman to whom he has not yet proposed. ii. The set S returned at termination is a perfect matching. b. Prove that binary search algorithm takes sub-linear time using asymptotic analysis. c. Let f and g be 2 functions that exists and is equal to $\lim_{n \rightarrow \infty} \left(\frac{f(n)}{g(n)} \right) = c$ some number $c > 0$. Then $f(n) = \Theta(g(n))$	05	Apply	CO1																																								
		05	Apply	CO2																																								

Course Outcomes meant to be assessed by the IA Test 1:

CO1: Define the basic concepts and analyze worst-case running times of algorithms using asymptotic analysis. (PO1, PO2, PO3, PSO2)

CO2: Illustrate the design techniques for graph traversal and divide and conquer algorithms and analyze their complexity by solving recurrence relations. (PO1, PO2, PO3, PSO2)